MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours) Examination – April 2026

Second Semester – MDC

Course Code: MG2MDCCFS102

Subject: Introduction to Computer Networks

QP Code: 25901649

Total Marks: 35

PART A Answer any 10 questions Each question carries 1 marks

1. The application layer provides network services directly to users, such as web browsing, email, and file transfer.

2. Network standards ensure compatibility between systems.

3. The physical layer is responsible for transmitting raw bits as electrical, optical, or radio signals over the communication medium.

4. Devices such as routers and gateways are responsible for connecting different networks.

5. Includes retransmission, window, acknowledgment policies.

6. The sender assumes the data was not received and retransmits the packet after a timeout.

7. Zone is a part of DNS namespace.

8. OSI means Open Systems Interconnection.

9. Passive attack only monitors data.

10. Access control prevents unauthorized access.

11. Ensures integrity of digital evidence.

12. Deadlock is indefinite waiting for resources.

PART B Answer any 3 questions. Each question carries 5 marks

13. Fiber optics are preferred in MAN because they provide very high bandwidth and support fast data transmission. They have low signal loss and are not affected by electromagnetic interference. This makes them reliable and suitable for long-distance communication.

14. SMTP is used to send emails between servers. It works over TCP and is mainly used for outgoing mail.

15. VRC uses parity bits, while LRC uses block checking. VRC is simple, whereas LRC is more reliable. LRC detects burst errors better.

16. Security mechanisms are methods used to protect data and systems from threats. They are classified into specific mechanisms (encryption, digital signature, access control, authentication) and pervasive mechanisms (auditing, security monitoring, recovery).

PART C Answer any 1 question. Each question carries 10 marks

17. Detailed explanation of Protocol & Standard

- Protocol: Rules of communication

- Standard: Agreed specifications

- Ensures interoperability

- Elements: Syntax, Semantics, Timing

- Examples: TCP/IP, HTTP

18. Detailed explanation of CRC with suitable example

- Cyclic Redundancy Check

- Uses polynomial division

- Remainder is added to data

- Receiver checks the remainder

- Zero remainder = No error

- Non-zero remainder = Error

- Detects burst errors